

IMMUNOBIOLOGY AND IMMUNOTHERAPY

High-avidity cathepsin-G-specific CAR T cells for the treatment of acute myeloid leukemia

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KEY POINTS

- CAR T cells targeting an HLA-A2-restricted epitope of CG show cytotoxicity against AML blasts without causing hematopoietic toxicity.
- LCK coexpressed in CG-specific CAR T cells enhances the functional avidity of the CAR T cells without modifying their specificity.

Chimeric antigen receptor (CAR) T cells specific for myeloid-associated antigens expressed on the cell surface of acute myeloid leukemia (AML) can cause depletion of normal myeloid progenitor cells. We developed a CAR specific for an HLA-A*02:01-restricted peptide of the myeloid-restricted cathepsin-G (CG) protein. CG-specific CAR T cells (CG1.CAR) were further engineered to increase their functional avidity. Specifically, we developed CG1.CAR T cells coexpressing the lymphocyte-specific protein tyrosine kinase (LCK) and duplicated CD3 ζ chain, which allows the functional recognition of the CG1 peptide as low as 0.025 μ M. Optimized CG1.CAR T cells displayed antileukemia effects in vitro and in vivo in patient-derived AML xenotransplant mouse models and did not cause hematopoietic toxicity in colony assays and humanized mice. Mechanistically, LCK overexpression in CG1.CAR T cells caused transcriptional modifications characterized by the overexpression of mitochondrial-encoded electron transport chain components that were correlated with increased mitochondrial mass and improved respiratory capacity. Based on these data, CG1.CAR T cells hold clinical potential for the treatment of AML.

Introduction

Chimeric antigen receptor (CAR) T cells targeting the CD19 antigen have significantly improved the clinical management of pediatric acute B-cell leukemia.¹ In sharp contrast, CAR T cells targeting acute myeloid leukemia (AML) have not yet made a clinical impact. One of the main concerns in developing safe and effective CAR T cells for AML is the lack of antigens highly expressed on the cell surface of leukemic cells, but absent in normal hematopoietic myeloid cells to prevent myeloablation.^{2,3} In contrast, intracytoplasmic proteins such as Wilms tumor suppressor gene (WT1)⁴ and myeloid serine protease proteinase 3 (PR3)⁵ can be targeted by CD8 T cells in the context of the major histocompatibility complex (MHC). In particular, the adoptive transfer of T cells engineered to express a high-affinity HLA-A*02:01 (HLA-A2)-restricted WT1 T-cell receptor (TCR) showed activity in patients with AML.⁶⁻⁸ However, TCR-engineered T cells demonstrated limited persistence after adoptive transfer, due, at least in part, to

inadequate T-cell costimulation, a feature that, in CAR T cells, is accomplished by the coexpression of costimulatory endodomains.^{1,9-12}

The development of monoclonal antibodies (mAbs) that specifically target MHC-restricted peptides challenged the dogma that peptides derived from intracytoplasmic proteins can only be targeted via TCR recognition. Two prototypes of such mAbs are the high affinity 8F4 mAb,¹³ which binds to the HLA-A2-restricted 9-mer PR1-peptide derived from PR3 and neutrophil elastase, and the ESK1¹⁴ mAb, which is specific for the RMF-peptide derived from the WT1 protein, also HLA-A2 restricted. Single chain variable fragments (scFvs) derived from the 8F4 and ESK1 mAbs have been used to generate CAR T cells and showed antileukemia activity.¹⁵⁻¹⁷

WT1 is highly expressed in AML but it is also expressed at low levels in early hematopoietic progenitors^{4,18,19} raising concerns for potential on-target off-tumor toxicity. In contrast, serine

proteases such as PR3 show more restricted expression in polymorphonuclear neutrophils under physiologic conditions and overexpression in AML.^{5,20,21} Cathepsin-G (CG) encoded by the CTSG gene is another serine protease that exerts proteolytic activity on cell debris, pathogens, and proteins^{22,23} and is normally stored within myeloid azurophil granules in granulocytes.²⁴ CG is broadly expressed in the cytosol of AML blasts and, in contrast to normal granulocytes, is ubiquitinated, processed, and presented in the MHC complex,²⁴ suggesting that CG-derived peptides can be considered pan-AML targets^{24,25} and targetable by scFvs specific for a peptide/MHC complex.

A critical aspect in developing scFv-based CAR T cells recognizing MHC-restricted peptides concerns their optimal functional avidity. T cells can effectively recognize their target epitope at the concentration of 5 μ M to 10 μ M,^{26,27} whereas scFv-based CAR molecules are 10- to 100-fold less sensitive than a TCR,²⁸⁻³¹ and may require up to a thousand surface antigen molecules to trigger functional signaling in T cells.³² This raises the concern that scFv-based CAR T cells may not sense low levels of the peptide in the MHC complex and trigger suboptimal T-cell signaling.^{33,34}

Here, we generated CG-specific CAR T cells characterized by high functional avidity and sensitive recognition of an HLA-A2–restricted CG peptide and potent antileukemia activity without causing hematopoietic toxicity.

Materials and methods

In vitro experiments with CAR T cells

Generation of retroviral supernatants and CAR T cells have been reported previously.^{35,36} Coculture and repeat stimulation coculture experiments of CAR T cells with tumor cells were performed as previously described.³⁶⁻³⁸ Coculture supernatant was collected 24 hours after each tumor stimulation to measure interferon gamma (IFN- γ) and interleukin-2 (IL-2) by enzyme-linked immunosorbent assay. For coculture experiments with primary AML samples obtained from HLA-A2⁺ patients with AML, peripheral blood mononuclear cells were reconstituted in StemSpan leukemic cell culture media (Stemcell Technologies, catalog no. 09605) in 12-well plates. Cells were incubated overnight at 37°C, and then AML blasts were counted and seeded in coculture at an effector-to-target ratio of 1:1 with CAR T cells or control T cells in the absence of cytokines, as previously described.³⁹ Coculture supernatants were collected at 24 hours to measure IFN- γ and IL-2 by enzyme-linked immunosorbent assay. On day 5, cells were collected and T cells and leukemic cells enumerated by flow cytometry. In other experiments, leukemic cells were viably cryopreserved at diagnosis, and autologous T cells were then obtained 21 days after induction chemotherapy to generate CAR T cells and coculture experiment in an autologous setting.

Xenograft, PDX mouse models, and humanized mice

Female 6- to 8-week-old NSG mice were purchased from the Animal Core Facility at University of North Carolina at Chapel Hill (UNC). Mice were housed in the Animal Core Facility at UNC. All mouse experiments were performed in accordance

with UNC animal husbandry and institutional animal care and use committee (IACUC) guidelines and were approved by UNC IACUC. Death of the animals was not allowed as the end point of the experiment and mice were euthanized when developing signs of discomfort as per protocol. NSG mice were injected IV via tail injection with OCI-AML2 tumor cells labeled with Firefly luciferase or AML patient-derived xenograft (PDX; CPDM_1060X, DFAM_22359, and DFAM_68555V3). After tumor engraftment, mice were randomized to receive non-transduced T cells or CAR T cells IV via tail injection. HLA-A2⁺ humanized mice were generated as previously described^{39,40} at the University of Maryland in accordance with Maryland animal husbandry and IACUC guidelines.

Statistical analysis

Paired or unpaired 2-tailed Student *t* tests and 1-way or 2-way analysis of variance were used to determine statistically significant differences between groups when the data passed the normality test; a nonparametric test was used to compare groups when normality assumptions could not be verified. Survival analysis was performed using the Kaplan-Meier method by log-rank test. All the data analyses were performed using GraphPad Software (version 10.1). Each dot represents 1 donor in the in vitro experiments and 1 mouse in the in vivo experiments. Each in vitro experiment was performed at least 3 times using different HLA-A2⁺ donors; data were summarized as the mean $\pm$ standard deviation or mean $\pm$ standard error of the mean as indicated. All *P* values < .05 were considered statistically significant.

Results

CG/HLA-A2–specific CAR (CG1.CAR) T cells target AML

The CG1 peptide (FLLPTGAEEA) derived from CG²⁴ was detected in 3 primary AML samples and in 3 AML cell lines using targeted parallel reaction monitoring–based nano–liquid chromatography analysis on immunopeptides isolated from HLA-A2–purified complexes. CG1 was detected at higher abundance than other peptides such as PR1 (VLQELNVTV, from PR3 and neutrophil elastase), CP1 (ALASLIRSV, from carboxypeptidase Q), and MP1 (RLFEQVMRI, from myeloperoxidase^{5,24,41}; Figure 1A-B; supplemental Figure 1, available on the *Blood* website). Using a publicly available AML RNA-sequencing (RNA-seq) data set from BeatAML, we analyzed CTSG mRNA expression in newly diagnosed AML, excluding acute promyelocytic leukemia.⁴² When compared with all samples, CTSG mRNA expression was lower in AML with CBFB::MYH11 fusion and was higher in AML with mutated nucleophosmin 1 (supplemental Figure 2A-B), potentially reflecting differences in CTSG mRNA expression at distinct myeloid differentiation stages. No significant differences were observed across other AML subtypes (supplemental Figure 2A-B). CTSG mRNA expression was also higher among female patients (supplemental Figure 2C), potentially reflecting a higher incidence of nucleophosmin 1–mutated AML among women.⁴³ No significant differences in CTSG mRNA expression were observed across the 2022 European LeukemiaNet AML risk categories (supplemental Figure 2D; European LeukemiaNet 2022 risk category annotations⁴⁴), between de novo vs secondary AML (supplemental Figure 2E),

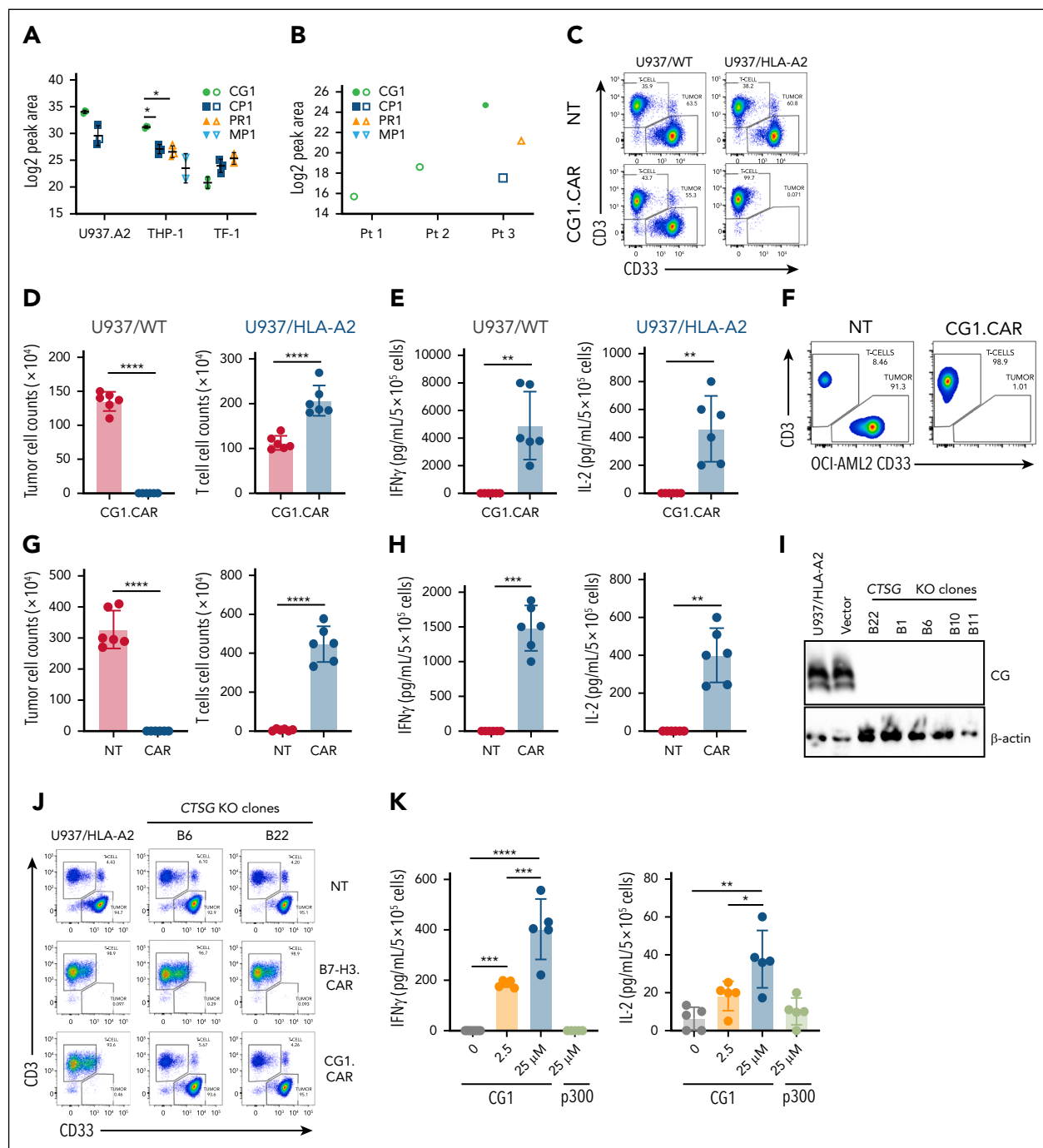


Figure 1. CG1.CAR T cells target AML. (A) Log₂-transformed peak areas of CG1, CP1, PR1, and MP1 peptides detected by targeted parallel reaction monitoring mass spectrometry in U937/HLA-A2, THP-1 and TF-1 AML cell lines. Peptides detected with high confidence (dotp > .90) are shown in solid symbols, and peptides detected with lower confidence (a dotp score of 0.75-0.90) are shown in open symbols. Error bars represent the mean ± standard deviation (SD); n = 3, *P < .5 as determined by log₂-transformed data and paired t test with Welch correction for U937/HLA-A2 cells and Welch analysis of variance (ANOVA) followed by the Dunnett T3 multiple comparisons test for THP-1 and TF-1 cells. (B) Log₂-transformed peak areas of CG1, CP1, and PR1 peptides detected from primary AML samples. Peptides detected with high confidence (dotp > .90) are depicted with solid symbols, and peptides detected with lower confidence (a dotp of 0.75 to 0.90) are depicted with open symbols. Cell numbers for patients 1, 2, and 3 were 72 × 10⁶, 116 × 10⁶, and 556 × 10⁶, respectively. (C) Representative flow cytometry plots of the coculture experiments in which U937/WT and U937/HLA-A2 cells were cocultured with CG1.CAR T cells or control NT cells at an effector-to-target ratio (E:T) of 1:1. All cells were collected on day 3 of coculture, and tumor cells and T cells were detected by CD33 and CD3 expression by flow cytometry, respectively. (D) Quantification of tumor cells and T cells in the coculture experiments described in panel C using counting beads; n = 6. (E) Quantification of IFN-γ and IL-2 in the supernatants collected at 24 hours in the coculture experiments described in panel C; n = 6. (F) Representative flow cytometry plots of the coculture experiments in which OCI-AML2 cells were cocultured with CG1.CAR T cells or control NT cells at an E:T ratio of 1:1 for 3 days. (G) Quantification of tumor cells and T cells in the coculture experiments described in panel F using counting beads; n = 6. (H) Quantification of IFN-γ and IL-2 in the supernatants collected at 24 hours in the coculture experiments described in panel F; n = 6. (I) Homozygous knockout of CTSG gene in the U937/HLA-A2 cell line by CRISPR/Cas-associated protein 9. CG protein was detected in cell clones obtained after CTSG knockout. U937/HLA-A2 and U937/HLA-A2 transfected with the control vector were used as controls. β-actin was used as loading control for the western blot. (J) Representative flow cytometry plots illustrating experiments in which U937/HLA-A2 cells and 2 CTSG knockout clones (B6 and B22) were cocultured with NT, B7-H3.CAR, or CG1.CAR T cells at an E:T ratio of 1:5. All cells were collected on day 4 of coculture,

response categories (supplemental Figure 2F), or among age categories (supplemental Figure 2G). There was also no significant difference in overall survival based on *CTSG* mRNA expression at diagnosis (supplemental Figure 2H). *CTSG* mRNA expression was also not significantly different between newly diagnosed patients and those with refractory or relapsed disease, although there was a trend toward lower expression at relapse (supplemental Figure 2I). Using a human E-ALPHA scFv library, 1008 phage clones were screened by flow cytometry against CG1 peptide-loaded T2 cells, and 5 scFvs (ET_S7, ET_S8, ET_S9, ET_S10, and ET_S11) were selected based on their binding to CG1 peptide-loaded T2 cells (supplemental Figure 3A). scFvs were cloned into a CAR format that included the human CD8 α stalk, CD28 endodomain, and CD3 ζ chain in retroviral vector,³⁵ and were successfully expressed in human T cells generated from HLA-A2⁺ healthy donors (supplemental Figure 3B). To ensure specificity, CAR T cells were cocultured with the myeloid leukemia U937/WT cell line (CG1⁺HLA-A2⁻) and the U937/HLA-A2 cell line (CG1⁺HLA-A2⁺). The melanoma M14/WT cell line (CG1⁺HLA-A2⁻) and the M14/HLA-A2 cell line (CG1⁺HLA-A2⁺) were used as additional controls. CAR T cells generated using the scFv ET_S8 and scFv ET_S10 showed no activity against the U937/WT, M14/WT, and M14/HLA-A2 cells, but activity against the U937/HLA-A2 cell line. In contrast, CAR T cells generated using the scFvs ET_S7, ET_S9, and ET_S11 showed cytotoxic activity against U937/HLA-A2 cells and both M14/WT and M14/HLA-A2 cells, indicating lack of stringent CG1/HLA-A2 specificity (supplemental Figure 3C-F). The CAR encoding the scFv ET_S10 was selected for further validation because T cells expressing this CAR demonstrated a trend of better expansion in vitro when tested in multiple HLA-A2⁺ donors compared with T cells expressing the CAR encoding the scFv ET_S8 (supplemental Figure 3G). Moreover, the scFv ET_S10 (from now on CG1.CAR) demonstrated stronger binding affinity in the single digit nanomolar range (supplemental Figure 3H). To further assess specificity, CG1.CAR T cells were tested against T2 cells loaded with 10 peptides representing the closest peptides to CG1 identified in the human proteome. No obvious cytokine release was observed (supplemental Figure 3I-J). Furthermore, CG1.CAR T cells did not show activation or IFN- γ release when tested against peripheral blood mononuclear cells obtained from healthy mismatched donors (supplemental Figure 3K-L). In coculture experiments, CG1.CAR T cells eliminated U937/HLA-A2 cells but not U937/WT cells and expanded in response to U937/HLA-A2 cells (Figure 1C-D). CG1.CAR T cells released both IFN- γ and IL-2 (Figure 1E), whereas control CD19.CAR T cells did not show antitumor activity and cytokine release (supplemental Figure 3M-N). Likewise, CG1.CAR T cells showed antitumor effects, proliferation, and cytokine release when cocultured with the OCI-AML2 cell line that physiologically expresses both CG1 and HLA-A2 (Figure 1F-H). *CTSG* knockdown in U937/HLA-A2 clones abolished the cytolytic activity of CG1.CAR T cells, whereas cell clones remained sensitive to the cytolytic effects of CAR T cells targeting another antigen³⁹ (Figure 1I-J).

CG1.CAR T cells showed a CG1-specific dose-response effect upon titration of the CG1 peptide (2.5 μ M and 25 μ M) in T2 cells (Figure 1K).⁴⁵ Overall, these data show that CG is broadly expressed in AML, the CG1 peptide is detectable by mass spectrometry in AML samples, and CG1.CAR T cells specifically target the CG1 peptide/HLA-A2 complex in a dose-dependent manner but produce relatively low amounts of cytokines.

CG1.CAR T cells overexpressing LCK and duplicated CD3 ζ chain show increased functional avidity

To enhance the functional avidity of CG1.CAR T cells, we generated CG1.CAR constructs in which we either duplicated the CD3 ζ intracytoplasmic tail domain⁴⁶ (CG1. $\zeta\zeta$.CAR) to increase the number of the tyrosine residues in the immunoreceptor tyrosine-based activation motifs or fused the CD3 ζ intracytoplasmic tail with either CD3 δ (CG1. $\zeta\delta$.CAR), CD3 γ (CG1. $\zeta\gamma$.CAR), or CD3 ϵ (CG1. $\zeta\epsilon$.CAR) intracytoplasmic tails (supplemental Figure 4A). All CAR molecules were expressed in T cells (supplemental Figure 4B), and CG1. $\zeta\zeta$.CAR T cells showed the highest functional avidity at the lowest (0.025 μ M) CG1 concentration (supplemental Figure 4C). LCK is the key kinase mediating the phosphorylation of the CAR CD3 ζ immunoreceptor tyrosine-based activation motifs.³⁶ We thus compared T cells expressing the CG1.CAR or CG1. $\zeta\zeta$.CAR with T cells cotransduced with a vector encoding LCK to obtain CG1.CAR + LCK or CG1. $\zeta\zeta$.CAR + LCK expressing T cells (Figure 2A-B). Both CG1. $\zeta\zeta$.CAR and CG1.CAR + LCK T cells showed superior functional avidity compared with the CG1.CAR T cells. However, CG1. $\zeta\zeta$.CAR + LCK T cells showed almost a fourfold increase in IFN- γ release at the lowest (0.025 μ M) CG1 concentration (Figure 2C). CG1.CAR and CG1. $\zeta\zeta$.CAR + LCK T cells showed similar basal CD3 ζ phosphorylation, but CG1. $\zeta\zeta$.CAR + LCK T cells showed increased CD3 ζ phosphorylation after CAR engagement (supplemental Figure 4D). No differences were observed in activation, proliferation, and apoptosis after CAR engagement between CG1.CAR and CG1. $\zeta\zeta$.CAR + LCK T cells (supplemental Figure 4E-G). However, CG1. $\zeta\zeta$.CAR + LCK T cells showed superior cytotoxic effects and lower TOX expression compared with CG1.CAR T cells after repeat stimulation with tumor cells (supplemental Figure 4H-I). CG1. $\zeta\zeta$.CAR T cells coexpressing LCK showed superior elimination of tumor cells in repeat stimulation experiments (Figure 2D; supplemental Figure 4J) and continued releasing IFN- γ and IL-2 after the third cycle of tumor stimulation (Figure 2E). CG1. $\zeta\zeta$.CAR + LCK T cells remained highly specific for the CG1 peptide/HLA-A2 complex because they did not target M14/HLA-A2 cells (HLA-A2⁺ but CG1⁻; supplemental Figure 4K) or other CG1-similar peptides (supplemental Figure 4L). In a xenograft OCI-AML2 tumor model in which tumor-bearing mice were treated with a sub-optimal dose of CAR T cells (1 $\times$ 10⁶ cells per mouse), T cells coexpressing CG1. $\zeta\zeta$.CAR and LCK showed superior antitumor effects, prolonged the probability of survival of mice (Figure 2F-G), and displayed superior content of T stem

Figure 1 (continued) and tumor cells and T cells were detected by CD33 and CD3 expression, respectively. (K) T2 cells were pulsed with the CG1 peptide at 2 different concentrations (2.5 μ M and 25 μ M) and incubated overnight at an E:T ratio of 1:1 with CG1.CAR T cells. IFN- γ and IL-2 were measured in the culture supernatant at 24 hours. T2 cells pulsed with an unrelated peptide (p300, HLA-A2 restricted PRAME peptide)⁴⁵ and T2 cells nonpulsed were used as controls; n = 5. **P* < .05; ***P* < .01; ****P* < .001; *****P* < .0001 by paired 2-tailed *t* test for panels D-E,G-H; 1-way ANOVA with Tukey multiple comparisons test for panel K. Data are shown as mean $\pm$ SD for panels D-E, G-H,K. CG1, peptide from CG; CP1, peptide from carboxypeptidase Q; MP1, peptide from myeloperoxidase; NT, nontransduced T cells; PR1, peptide from PR3 and neutrophil elastase; Pt, patient.

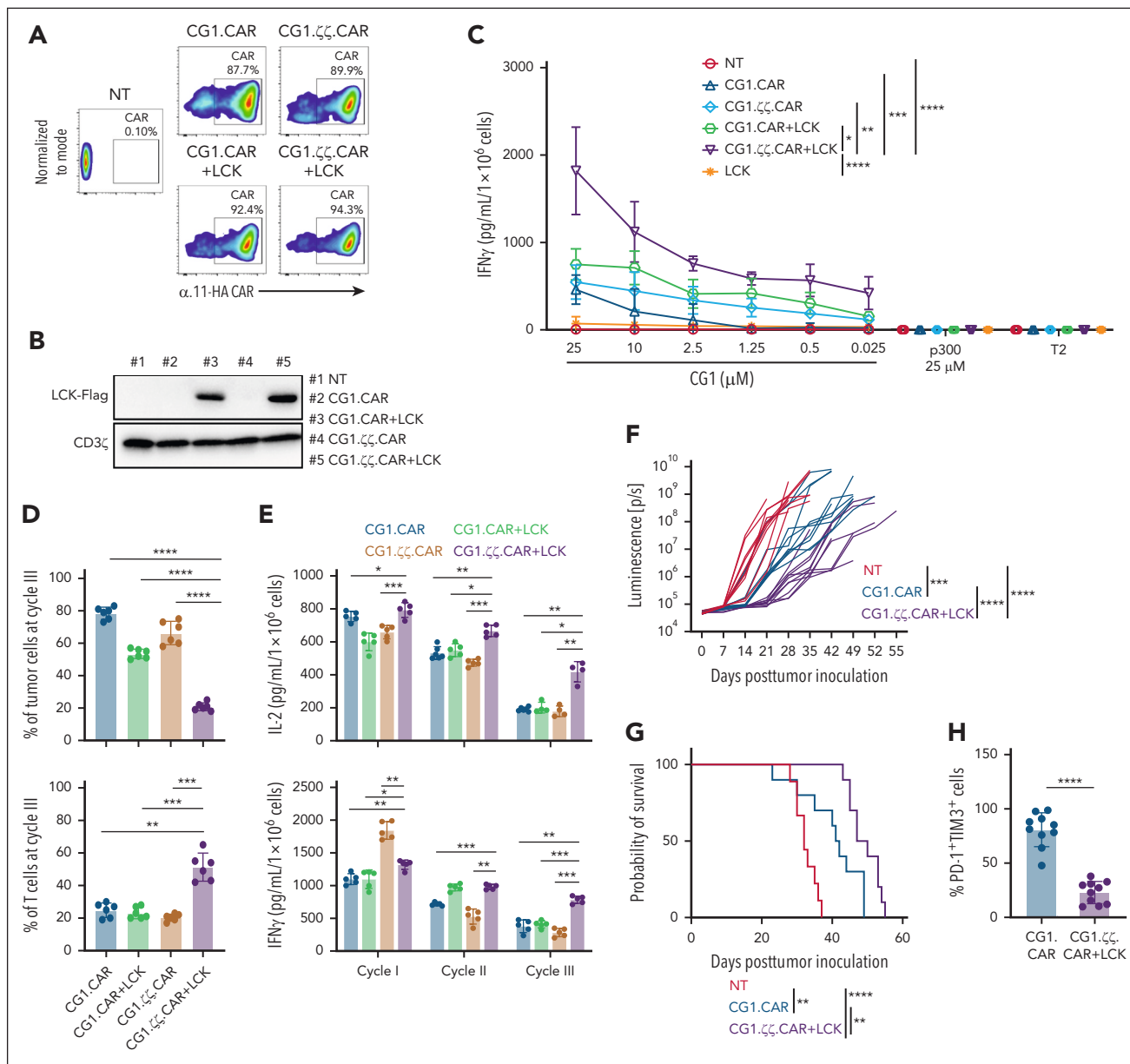


Figure 2. CD3 ζ duplication and LCK overexpression increase the functional avidity of CG1.CAR T cells. (A-B) Representative flow cytometry plots showing CAR expression (A) and western blot showing LCK transgenic expression (B) in T cells transduced with vectors encoding the CAR alone (CG1.CAR and CG1.ζζ.CAR) and in T cells cotransduced with the vectors encoding the CAR and LCK (CG1.CAR + LCK and CG1.ζζ.CAR + LCK). CAR expression was detected using the α.11-HA antibody (Ab), whereas LCK was detected using a FLAG-Tag Ab. (C) Quantification of IFN-γ in the supernatants collected at 24 hours in coculture experiments in which CG1.CAR, CG1.ζζ.CAR, CG1.CAR + LCK, and CG1.ζζ.CAR + LCK T cells were cocultured with T2 cells pulsed with the CG1 peptide at different concentrations ranging from 0.025 μM to 25 μM. LCK indicates T cells transduced only with the vector encoding LCK. T2 cells nonpulsed and T2 cells pulsed with the p300 peptide at 25 μM were used as controls; n = 7. (D) Quantification of tumor cells and T cells in repeat stimulation coculture experiments in which CG1.CAR, CG1.CAR + LCK, CG1.ζζ.CAR, and CG1.ζζ.CAR + LCK T cells were cocultured with OCI-AML2 cells added every 3 days for 3 consecutive rounds. At the end of the third cycle, all cells were collected, and tumor cells and T cells were detected by CD33 and CD3 expression by flow cytometry, respectively; n = 6. (E) Quantification of IL-2 and IFN-γ in the supernatant collected at 24 hours in the repetitive coculture experiments described in panel D; n = 4-6. (F) Tumor growth kinetics measured by bioluminescence in NSG mice inoculated with OCI-AML2 tumor cells (0.8×10^6 cells per mouse) by tail vein injection and treated at day 4 with either NT, CG1.CAR, or CG1.ζζ.CAR + LCK T cells (1×10^6 cells per mouse); n = 10 mice per group. Data summarize 2 independent experiments. (G) Kaplan-Meier survival curve of the experiments described in panel F. (H) Phenotypic characterization of T cells in the bone marrow of mice at the time of euthanasia of the model described in panel F; n = 10 mice per group. * $P < .05$; ** $P < .01$; *** $P < .001$; **** $P < .0001$ as determined by 1-way ANOVA with Tukey multiple comparisons test at the peptide concentration 0.025 μM (C), 1-way ANOVA with Dunnett multiple comparisons test for panel D, 2-way ANOVA with Dunnett multiple comparisons test for panel E, log-transformed 1-way ANOVA with Tukey comparisons test at day 21 (F), log-rank test for panel G, and unpaired 2-tailed t test for panel H. Data are shown as mean $\pm$ SD for panels C-E, H. NT, nontransduced T cells.

memory-like cells in vivo when assessing CD45RA and CCR7 expression as markers of memory cells^{47,48} (supplemental Figure 4M), and T cells showed lower coexpression of programmed cell death protein 1 (PD1) and T-cell immunoglobulin

and mucin-domain containing-3 (TIM3; Figure 2H). In another independent experiment in which mice were treated with a higher dose of CAR T cells (6×10^6 cells per mouse), CG1.ζζ.CAR + LCK T cells showed superior expansion in the

bone marrow at day 45 (supplemental Figure 4N). However, tumor relapses were still observed, and tumor analysis showed HLA-A2 loss in OCI-AML2 cells (supplemental Figure 4O). To facilitate the clinical translation of the proposed approach, we then generated a single vector encoding both CG1.ζζ.CAR and LCK (CG1.ζζ.CAR.LCK). Comparative experiments *in vitro* and *in vivo* showed that CG1.ζζ.CAR.LCK T cells retained the appropriate functional avidity and antitumor effects and the single vector was used for further characterization of CG1.CAR T cells (supplemental Figure 5). Overall, these results indicate that LCK overexpression combined with higher number of tyrosine residues available for phosphorylation confers higher functional avidity to the CAR targeting the CG1 peptide–HLA-A2 complex without compromising their specificity.

LCK overexpression promotes enhanced activation and oxidative metabolism in CG1.CAR T cells

To investigate the influence of LCK overexpression in CAR T cells, we performed RNA-seq analysis of CG1.CAR and CG1.ζζ.CAR.LCK T cells before antigen stimulation, 24 hours after antigen stimulation, and 5 days after antigen stimulation⁴⁹ (Figure 3A). Both CG1.CAR and CG1.ζζ.CAR.LCK T cells showed a molecular signature compatible with activation at day 1 after antigen stimulation (Figure 3B–C; supplemental Figure 6A), with limited differences in gene expression (Figure 3D). In contrast, at day 5 after stimulation, a suite of genes that encode mitochondrial electron transport chain components contained in Complex I (MT-ND1–ND6), III (MT-CytB), IV (MT-COI–COIII), and V (MT-ATP6, MT-ATP8), were enriched in CG1.ζζ.CAR.LCK T cells (Figure 3E; supplemental Figure 6B–C). Seahorse mitochondrial stress test showed that CG1.ζζ.CAR.LCK T cells displayed increased oxygen consumption rate (OCR) and improved reserve respiratory capacity compared to CG1.CAR T cells (Figure 3F). Furthermore, CG1.ζζ.CAR.LCK T cells showed improved metabolic plasticity marked by increased extracellular acidification rate in the glucose stress test (Figure 3F; supplemental Figure 6D–E), improved mitochondrial membrane potential as measured by tetramethylrhodamine and increased mitochondrial mass (MitoTracker Green) on a per cell basis compared to CG1.CAR T cells (Figure 3G). Confocal microscopy confirmed increased mitochondria per cell in CG1.ζζ.CAR.LCK T cells after stimulation compared to CG1.CAR T cells (Figure 3H–I). Overall, these results indicate that LCK overexpression confers exquisite transcriptional changes in a set of genes conferring metabolic imprinting for enriched mitochondrial function.

CG1.ζζ.CAR.LCK T cells do not target normal myeloid progenitors

We compared CTSG mRNA expression in AML samples vs healthy hematopoietic progenitors using publicly available RNA-seq and single cell RNA-seq data sets.^{50,51} We found higher CTSG mRNA expression in healthy pooled CD34⁺ cells and healthy individual bone marrow mononuclear cells, and higher CTSG mRNA expression in Lin[−]CD34⁺CD164^{low} progenitors (Figure 4A–B). CG protein expression was confirmed in CD34⁺ selected hematopoietic stem cells obtained from umbilical cord blood units using confocal microscopy (Figure 4C). We then performed standard colony-forming unit assays in which HLA-A2⁺ or HLA-A2[−] umbilical cord blood–derived CD34⁺ cells were

exposed to either CG1.CAR or CG1.ζζ.CAR.LCK T cells generated from healthy HLA-A2⁺ donors and then plated to form colony-forming units. No significant differences in colony formation were observed between HLA-A2⁺ or HLA-A2[−] progenitors cocultured with either control T cells or CG1.CAR T cells (Figure 4D). Similarly, we did not observe differences in cell viability or IFN-γ release when HLA-A2⁺ or HLA-A2[−] CD34⁺ cells were cocultured in liquid cultures with either control T cells or CG1.CAR T cells (Figure 4E–G). We finally evaluated the toxicity of CG1.CAR T cells in HLA-A2⁺ humanized mice³⁹ (Figure 4H). Human T cells were collected from the spleen of humanized mice and were used to generate control T cells and CG1.ζζ.CAR.LCK T cells. These cells were infused in HLA-A2⁺ humanized mice of the same cohort of mice from which T cells were originally generated. Upon treatment, mice did not show signs of toxicities measured as body weight loss (Figure 4I). We also did not observe differences in the cellular composition of the bone marrow (Figure 4J–K), peripheral blood, and spleen (supplemental Figure 7) at day 10 after treatment in the first cohort of mice and at day 21 after treatment in the second cohort of mice. Overall, these results indicate that CG1.ζζ.CAR.LCK T cells do not cause toxic effects on normal hematopoietic progenitor cells.

CG1.ζζ.CAR.LCK T cells target primary AML *in vitro* and *in vivo*

We quantified CTSG mRNA expression by quantitative polymerase chain reaction in 10 deidentified HLA-A2⁺ adult AML samples at the time of diagnosis (Figure 5A). We then enumerated CD45⁺CD34⁺ and CD45⁺CD123⁺ cells by flow cytometry in these samples (Figure 5B–C), and found a correlation between CTSG mRNA expression levels and the percentage of CD45⁺CD34⁺ cells (Figure 5D). We also evaluated CG protein expression by western blot on total cell lysates, and observed a correlation between protein and mRNA expression (Figure 5E). In 10 additional AML samples we performed immunofluorescence analysis and confocal microscopy analysis, which confirmed CG expression mostly in the cytoplasm of the cells (Figure 5F; supplemental Figure 8A). Samples 1, 2, 5, 7, 8, and 9 were available in sufficient amounts to be tested in coculture experiments with control T cells, CG1.CAR T cells, and CG1.ζζ.CAR.LCK T cells generated from healthy HLA-A2⁺ donors. CG1.CAR and CG1.ζζ.CAR.LCK T cells eliminated AML blasts irrespective of the level of CG1 expression (Figure 5G–H), and released IFN-γ and IL-2 (Figure 5I–J). The antitumor effects of CG1.CAR and CG1.ζζ.CAR.LCK T cells were then independently validated in 5 additional adult and pediatric AML samples in an autologous setting. In these experiments, autologous CAR T cells expressing either CG1.CAR or CG1.ζζ.CAR.LCK constructs showed cytotoxic activity and cytokine release against autologous AML blasts (Figure 5K–L; supplemental Figure 8B–H).

To test the efficacy of CG1.ζζ.CAR.LCK T cells *in vivo*, we used 3 HLA-A2⁺ AML PDX models (CPDM_1060X, DFAM_22359, and DFAM_68555V3). PDXs were inoculated via tail IV in NSG mice, and tumor engraftment was monitored in the peripheral blood by flow cytometry to detect CD123 and CD33. When blast counts reached 10% to 20% in the peripheral blood (day 21), mice were randomized to receive either control T cells, CG1.CAR T cells, or CG1.CAR.ζζ.LCK T cells (5 × 10⁶ cells per

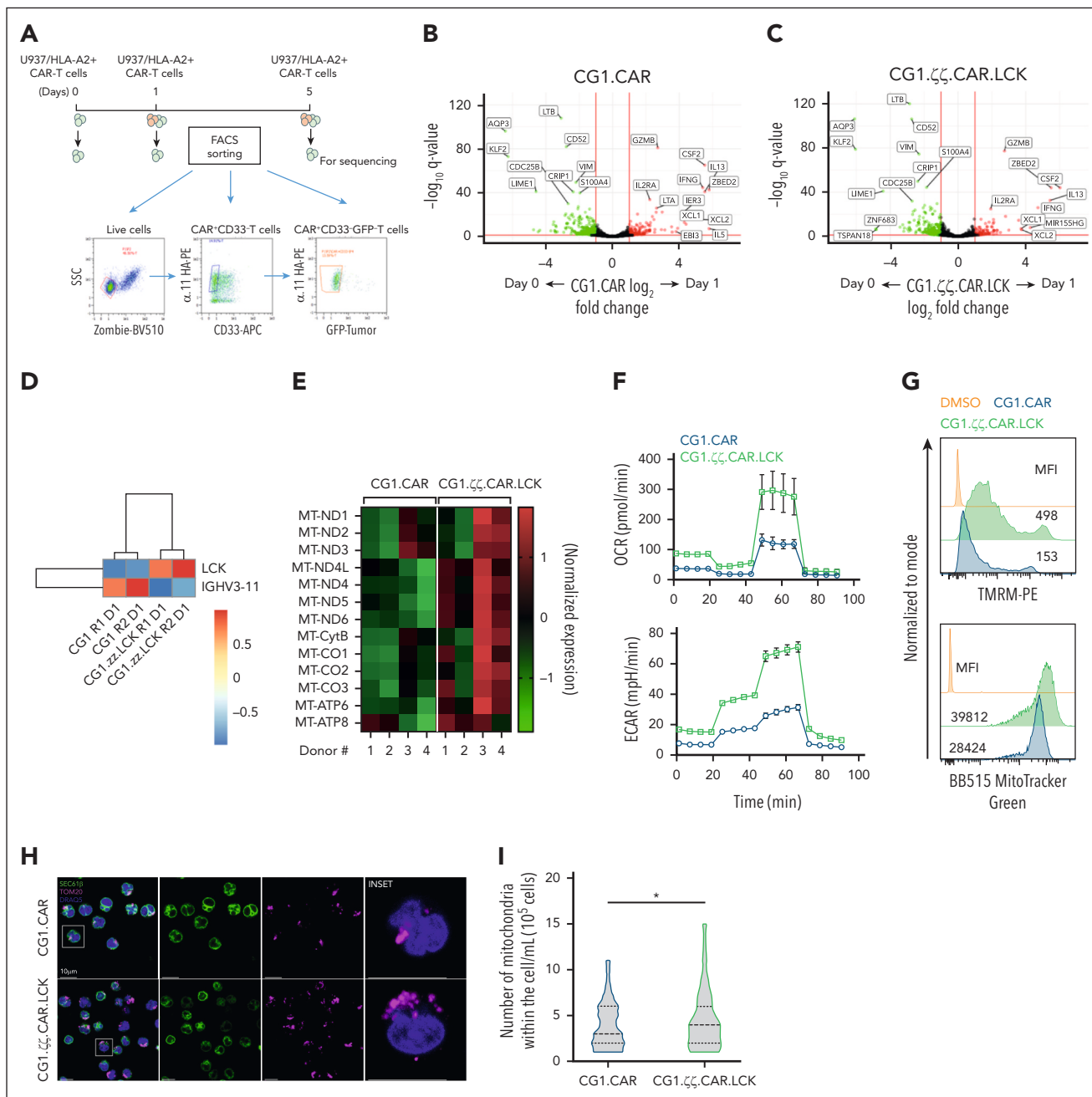


Figure 3. CG1.ζζ.CAR- and LCK-expressing T cells show increased metabolic activity. (A) Schematic of the sorting strategy of CG1.CAR and CG1.ζζ.CAR.LCK T cells cocultured with U937/HLA-A2 cells. (B-C) Differentially expressed genes in CG1.CAR T cells (B) and CG1.ζζ.CAR.LCK T cells (C) before stimulation (day 0) and at day 1 after stimulation. (D) Heat map showing genes differentially expressed at day 1 after stimulation in CG1.CAR and CG1.ζζ.CAR.LCK T cells. (E) Heat map illustrating a suite of genes that encode mitochondrial electron transport chain (ETC) components that are enriched in CG1.ζζ.CAR.LCK T cells compared with CG1.CAR T cells 5 days after stimulation. (F) Representative plot showing the ECAR and OCR in T cells expressing either the CG1.CAR or CG1.ζζ.CAR.LCK upon stimulation with tumor cells; n = 6. (G) Representative flow plots illustrating the quantification of active mitochondria with intact membrane potentials and active mitochondria using TMRM mitochondria probe and MitoTracker Green staining, respectively, in T cells expressing either the CG1.CAR or CG1.ζζ.CAR.LCK and cocultured with tumor cells at an E:T ratio of 2:1; n = 6. (H-I) Representative confocal microscopy (H) and quantification (I) of CG1.CAR and CG1.ζζ.CAR.LCK CD8 T cells selected after coculture with tumor cells at the E:T ratio of 2:1 for 2 days. Sec61p stains the endoplasmic reticulum, whereas TOM20 and DRAQ5 stain mitochondria and nucleus, respectively; n = 6. 1-sided paired t test for panel E; 12 of 13 ETC genes achieved $P < .05$ with ND3 $P = .066$; * $P < .05$ by paired 2-tailed t test for panel I. DMSO, dimethyl sulfoxide; ECAR, extracellular acidification rate; MFI, mean fluorescence intensity; OCR, oxygen consumption rate; PE, phycoerythrin; TMRM, tetramethylrhodamine.

mouse) generated from HLA-A2⁺ healthy donors. Upon treatment, mice did not show signs of toxicity measured as body weight loss (supplemental Figure 9A). In these experiments, death of the animal was not allowed as end point of the experiment and thus mice were euthanized between days 14

and 29 after treatment to assess the tumor burden. In the PDX CPDM_1060X model (Figure 6A), mice were euthanized on day 28 after treatment (day 49), and the bone marrow and spleens were collected for analysis. We observed higher tumor burden in the bone marrow of mice treated with CG1.CAR T cells

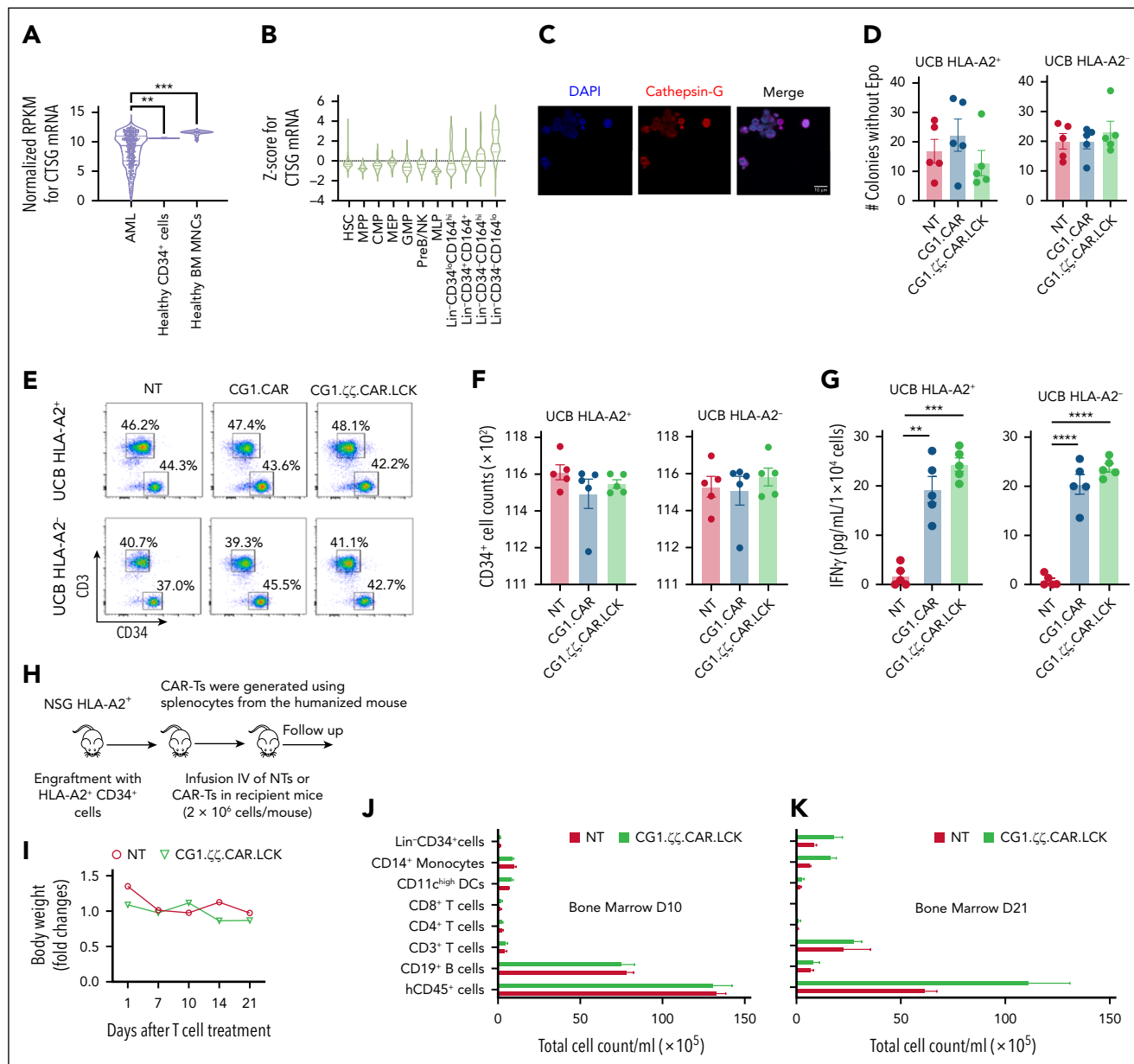


Figure 4. CG1.ζζ.CAR.LCK T cells do not target normal myeloid progenitors in vitro and in vivo in a humanized mouse model. (A) CTSG mRNA expression in AML samples, pooled CD34⁺ cells and BM MNCs from healthy donors from the BeatAML1 cohort.⁴⁰ Data obtained from <http://www.vizome.org>. The solid line indicates the median, and dotted lines indicate the 25th and 75th quartiles. (B) Comparison of CTSG mRNA expression on hematopoietic progenitors from single-cell RNA-seq data set of sorted BM lineage-negative hematopoietic progenitors.⁴⁵ Data obtained from <https://bloodspot.eu>. (C) Representative confocal image (63x) of CG expression in CD34⁺ cells pooled from 6 umbilical cord blood (UCB) units. Scale bars, 10 μm. (D) UCB-derived CD34⁺ cells obtained from either HLA-A2⁺ or HLA-A2⁻ samples were cocultured for 24 hours with CG1.CAR- or CG1.ζζ.CAR.LCK-expressing T cells or control NT T cells obtained from healthy HLA-A2⁺ donors at an E:T ratio of 10:1. Cells were then transferred to a semisolid medium and cultured for 2 weeks. Hematopoietic colonies were enumerated in a double-blind fashion by 2 individuals; n = 5; P > .05. (E) Representative flow cytometry plots of coculture experiments in which HLA-A2⁺ and HLA-A2⁻ UCB-derived CD34⁺ cells were cultured with CG1.CAR- or CG1.ζζ.CAR.LCK-expressing T cells or control NT T cells at an E:T ratio of 1:1. All cells were collected on day 5 of coculture, and stem cells and T cells were detected by CD34 and CD3 expression by flow cytometry, respectively. (F) CD34⁺ cell counts for HLA-A2⁺ and HLA-A2⁻ UCB-derived CD34⁺ cells cocultured with T cells as described in panel B and determined by counting beads by flow cytometry; n = 5, P > .05. (G) Quantification of the IFN-γ in the supernatants collected at 24 hours in the coculture experiments illustrated in panel D; n = 5. (H) Schematic of the HLA-A2 humanized mouse model. Control and CG1.ζζ.CAR.LCK T cells were generated by isolating human T cells from the spleen of a mouse of the cohort of mice engrafted with CD34⁺ cells. Control NT or CG1.ζζ.CAR.LCK T cells were then inoculated in humanized mice of the same cohort. (I) Measurement of the body weight of the humanized mice described in panel H. Summary of 2 independent experiments with 2 different cohorts of humanized mice engrafted with 2 different preparations of CD34⁺ cells. (J-K) BM was collected on day 10 in the first cohort of mice and on day 21 in the second cohort of mice from 1 femur. BM composition was defined by flow cytometry; n = 3 to 4 mice per group; P > .05. **P < .01; ***P < .001; ****P < .0001 as determined by 1-way ANOVA followed by Šidák multiple comparisons test for panel A, 1-way ANOVA with Tukey multiple comparisons test for panels D, G, Friedman test with Dunnett multiple comparisons test for panel F, 2-way ANOVA with Šidák multiple comparisons test for panels J-K. Data are shown as mean ± standard error of the mean (SEM) for panels D, F, G, J-K. BM, bone marrow; DAPI, 4',6-diamidino-2-phenylindole; MNCs, mononuclear cells; NT, nontransduced T cells.

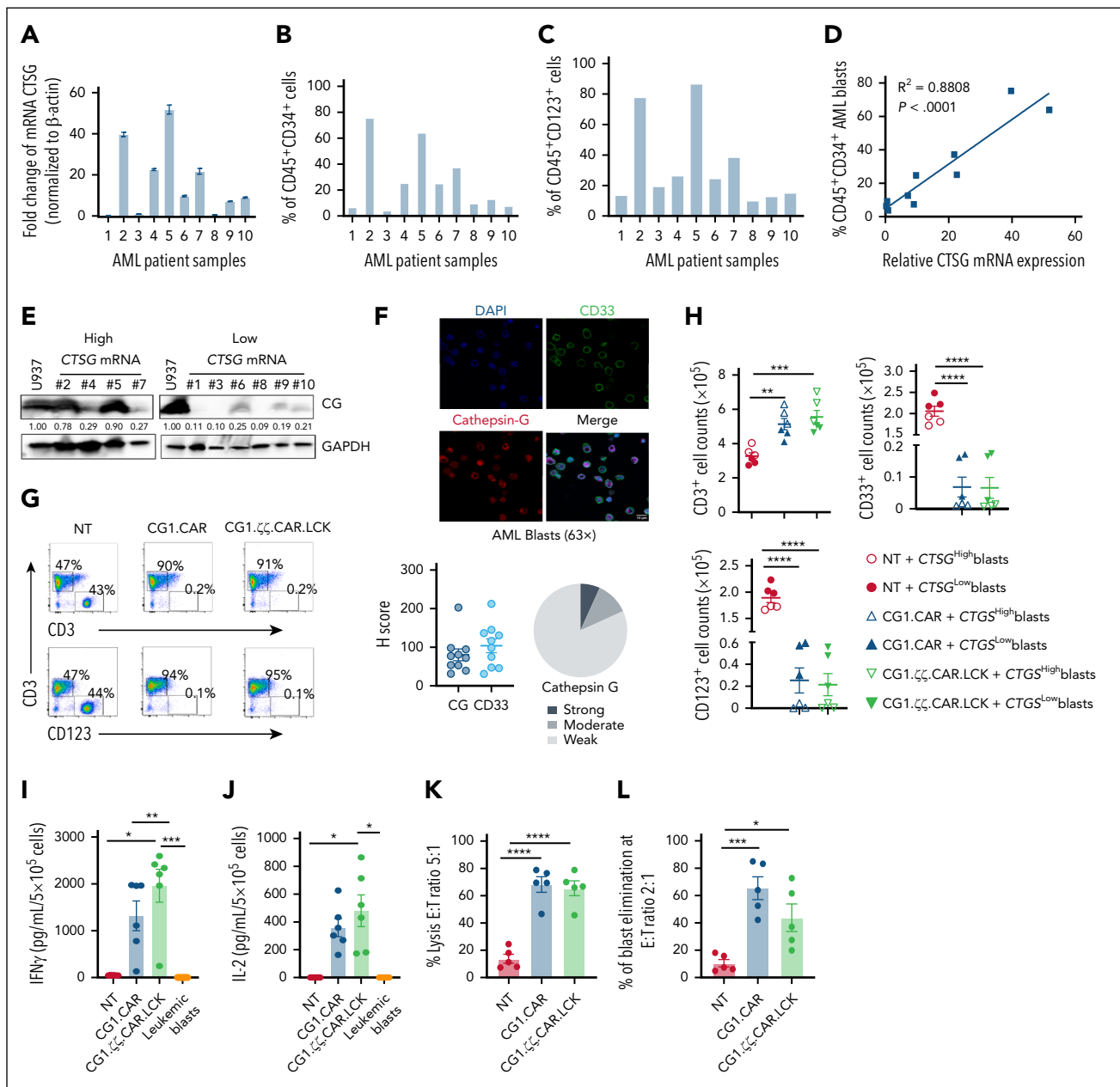


Figure 5. CG1.ζζ.CAR.LCK T cells target primary AML in vitro. (A) CTSG mRNA expression in primary AML samples obtained from adult HLA-A2⁺ patients at UNC. mRNA expression was determined by quantitative polymerase chain reaction (qPCR) using mRNA extracted from total peripheral blood mononuclear cell and CTSG mRNA expression was normalized to β-actin. (B-C) Flow cytometry analysis showing the percentage of CD45⁺CD34⁺ (B) and CD45⁺CD123⁺ (C) cells in AML samples described in panel A. (D) Linear regression of CD45⁺CD34⁺ cells and CTSG mRNA expression in AML samples described in panel A. (E) CG protein expression was determined by western blot in AML samples described in panel A. AML samples were divided into CTSG^{High} mRNA- and CTSG^{Low} mRNA-expressing samples based on qPCR values. Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) expression was used as a loading control for the western blot. U937 cell lysate was used as a positive control. Densitometric values are displayed and represent CG protein band intensities normalized to their respective GAPDH loading control, and are expressed as a fold change relative to the U937 control for each gel. (F) Representative confocal image (63×) of CG and CD33 expression in AML; scale bars, 10 μm (upper panel); H score of CG and CD33 expression based on immunofluorescence in 10 AML samples (lower panels). (G) Representative flow cytometry plots of the experiments in which control NT, CG1.CAR, and CG1.ζζ.CAR.LCK T cells generated from healthy HLA-A2⁺ donors were cocultured with the AML sample 2 (CTSG^{High}). At day 4 of coculture, all cells were collected to enumerate CD3⁺, CD33⁺, and CD123⁺ cells by flow cytometry. (H) Quantification of CD3⁺ cells, CD33⁺ cells, and CD123⁺ cells in the coculture experiments described in panel G with AML samples 1, 2, 5, 7, 8, and 9 that were available in sufficient amount to be tested in coculture experiments. Samples CTSG^{Low} and CTSG^{High} are indicated with solid and open shapes, respectively; n = 6. (I-J) Quantification of IFN-γ (I) and IL-2 (J) in the coculture supernatant collected at 24 hours in the coculture experiments with the AML samples 1, 2, 5, 7, 8, and 9; n = 6. (K) Cytotoxic assay at the E:T ratio of 5:1 of autologous CG1.CAR and CG1.ζζ.CAR.LCK T cells against autologous primary AML blasts; n = 5. (L) Coculture experiment in which AML blasts described in panel L were cocultured with either CG1.CAR or CG1.ζζ.CAR.LCK autologous T cells at an E:T ratio of 2:1. At day 4 of coculture all cells were collected to enumerate CD3⁺ and CD33⁺ cells by flow cytometry; n = 5. *P < .05; **P < .01; ***P < .001; ****P < .0001 by 1-way ANOVA with Tukey multiple comparisons test for panel H; Friedman test with Dunnett multiple comparisons test for panels I-J; 1-way ANOVA with Tukey multiple comparisons test for panels K-L. Data are shown as mean ± SEM for all panels. DAPI, 4',6-diamidino-2-phenylindole; NT, nontransduced T cells.

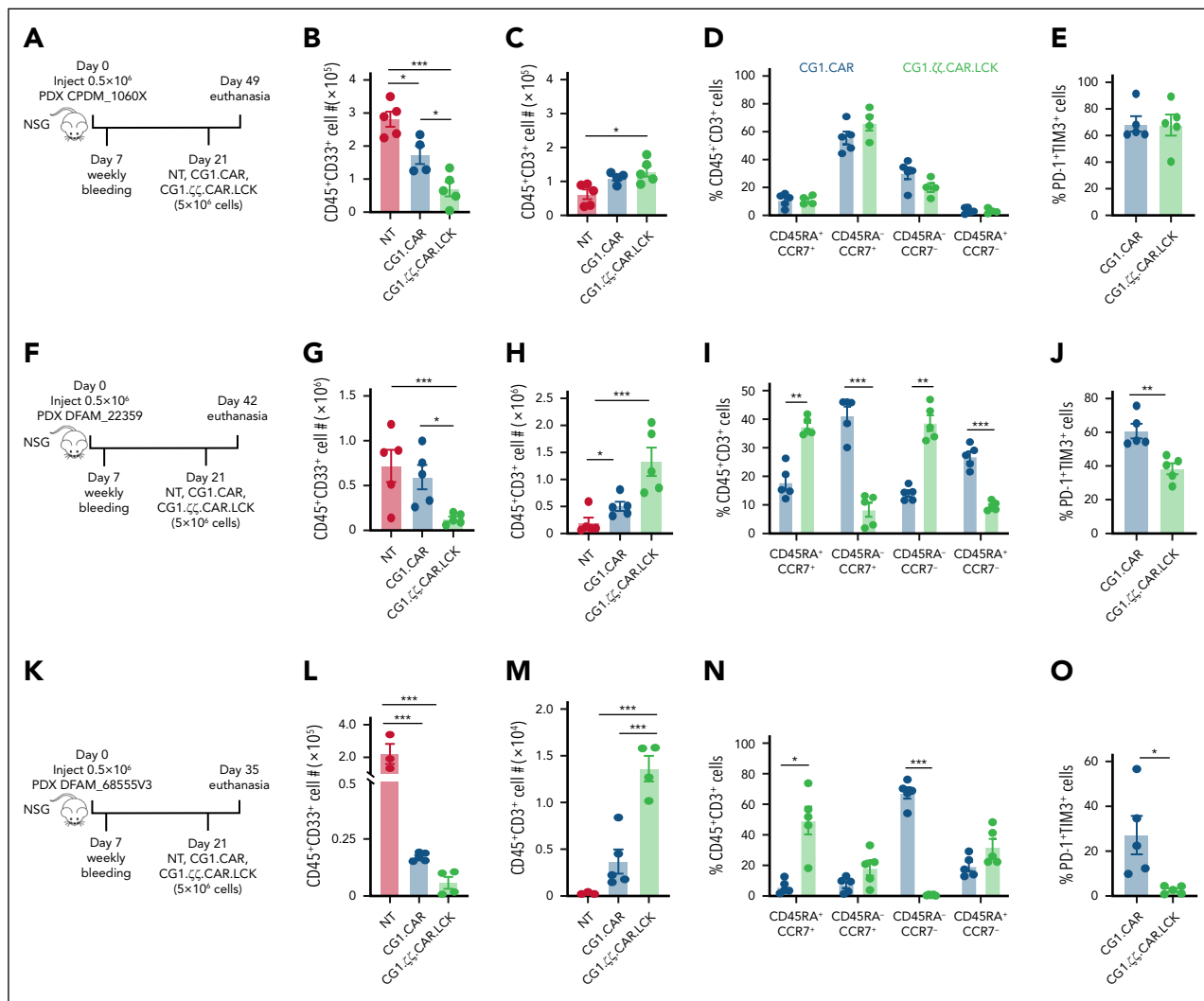


Figure 6. CG1.ζζ.CAR.LCK T cells display antitumor activity in PDX models of AML. (A-E) Schematic of the CPDM_1060X PDX model (A). All mice were euthanized at day 49, and tumor cells (CD45⁺CD33⁺) (B) and T cells (CD45⁺CD3⁺) (C) were quantified in the bone marrow by flow cytometry. (D-E) T cells were phenotypically characterized based on CD45RA and CCR7 expression (D) and PD-1 and TIM3 coexpression (E); n = 5 mice per group. (F-J) Schematic of the DFAM_22359 PDX model (F). All mice were euthanized on day 42, and tumor cells (CD45⁺CD33⁺) (G) and T cells (CD45⁺CD3⁺) (H) were quantified in the bone marrow by flow cytometry. (I-J) T cells were phenotypically characterized based on CD45RA and CCR7 expression (I) and PD-1 and TIM3 coexpression (J); n = 5 mice per group. (K-O) Schematic of the DFAM_68555-V3 PDX model (K). All mice were euthanized at day 35, and tumor cells (CD45⁺CD33⁺) (L) and T cells (CD45⁺CD3⁺) (M) were quantified in the bone marrow by flow cytometry. (N-O) T cells were phenotypically characterized based on CD45RA and CCR7 and (N) and PD-1 and TIM3 coexpression (O); n = 3 to 5 mice per group. **P* < .05; ***P* < .01; ****P* < .001 by 1-way ANOVA with Tukey multiple comparisons test for panels B-C, G-H, L-M; 2-way ANOVA with Šidák multiple comparisons test for panels D, I, N; and unpaired 2-tailed *t* test for panels E, J, O. Data are shown as mean ± SEM for all panels.

compared with mice treated with CG1.ζζ.CAR.LCK T cells (Figure 6B), without significant differences in T-cell counts and their phenotypic composition (Figure 6C-E). In the second PDX DFAM_22359 model (Figure 6F), mice were euthanized on day 21 after treatment (day 42). In this model, we confirmed the superior antitumor effect of CG1.ζζ.CAR.LCK T cells in the bone marrow compared with CG1.CAR T cells (Figure 6G). Additionally, in this model, we observed a significant increase in T-cell counts (Figure 6H), enrichment in T stem memory-like cells CD45RA⁺CCR7⁺ (Figure 6I), and reduction of PD1⁺TIM3⁺ exhausted T cells in mice treated with CG1.ζζ.CAR.LCK T cells compared with CG1.CAR T cells (Figure 6J). In the third DFAM_68555V3 PDX model (Figure 6K), we confirmed on day 14 after treatment (day 35) the superior antitumor effects of CG1.ζζ.CAR.LCK T cells in the bone marrow compared

with CG1.CAR T cells (Figure 6L), as well as superior T-cell counts (Figure 6M), and superior preservation of CD45RA⁺CCR7⁺ cells (Figure 6N) with reduced coexpression of PD1 and TIM3 (Figure 6O). In all 3 PDX models, the analysis performed in the bone marrow was corroborated by similar results in the spleen (supplemental Figure 9B). Overall, these results indicate that CG1.ζζ.CAR.LCK T cells show antitumor activity in AML PDX models.

Discussion

On-target, off-tumor toxicity remains one of the major challenges in the development of CAR T cells in AML. Here, we propose that CAR T cells targeting an HLA-A2–restricted

peptide derived from the myeloid-restricted CG protein are safe and effective in clinically relevant AML preclinical models.

Here, we confirmed high expression of *CTSG* mRNA in all morphologic subtypes of AML, and demonstrated that the HLA-A2–restricted CG1 peptide (FLLPTGAEA)²⁴ can be detected by mass spectrometry in primary AML blasts more abundantly than peptides of other myeloid-restricted proteases such as MPO, neutrophil elastase, and PR3.⁵ We have detected CG protein by western blot in AML blasts and cytoplasmic localization of the protein using immunofluorescence and confocal microscopy. We finally demonstrated that CAR T cells targeting the CG1/HLA-A2 complex eliminate leukemic cells both in vitro and in vivo.

Using available RNA-seq data sets, we compared *CTSG* mRNA expression in AML vs hematopoietic progenitors.^{50,51} We observed that *CTSG* mRNA is expressed in healthy CD34⁺ cells with the highest expression in Lin[−]CD34⁺CD164^{low} progenitors. We have also found that CG can be detected in CD34⁺ cells obtained from UCB units using confocal microscopy. Despite mRNA and protein expression, our functional data show that CG1.CAR T cells do not cause toxicity to hematopoietic progenitors in vitro or in humanized mice. The lack of toxicity of CG1.CAR T cells in normal hematopoietic progenitors could be attributed to several factors. It was previously reported that Lin[−]CD34⁺CD38[−] cells isolated from AML samples show higher expression of CG protein than leukemic cells.²⁴ Furthermore, CG in leukemic cells, but not in normal cells, is ubiquitinated, allowing its degradation and MHC presentation.^{24,25} It was also reported that hematopoietic progenitors are less sensitive to CAR T cell–mediated cytotoxicity due, at least in part, to the expression of granzyme-B inactivating serine proteases.⁵² Despite these indications, the initial clinical translation of CG1.CAR T cells in patients with relapsed AML may be indicated in patients with an available bone marrow transplant donor. Alternatively, the inclusion of a clinically validated safety switch gene can be considered to selectively ablate CAR T cells in case of myeloid suppression or on-target off-tumor toxicities.^{53,54} Finally, our preclinical data also indicate that AML tumor cells can lose or downregulate HLA under selective pressure of CG1.CAR T cells, suggesting that this therapeutic approach may be more effective as bridge therapy to an allogeneic stem cell transplant.

CARs require up to a thousand surface antigen molecules to trigger the downstream signaling compared with a conventional TCR.^{28–32} For conventional CARs, the target antigen density threshold is ~3000 copies per cell.⁵⁵ Although some peptides are presented in high copy numbers, the antigen density for most peptide-MHC complexes is at ~100 copies per cell, which is far lower for CARs without enhancement.⁵⁶ This weakness raises the concern that scFv-based targeting of an MHC-restricted peptide may not sense low levels of the peptides causing insufficient activation of T cells.^{33,34} We have previously reported that LCK plays a critical role in initiating the CAR signaling.³⁶ Here, we show that LCK overexpression can enhance the functional avidity of the CG1/HLA-A2–specific CAR promoting superior antitumor effects. It remains to be explored whether LCK overexpression enhances the functional avidity of conventional CARs targeting surface proteins. Finally,

LCK overexpressing CAR T cells did not show autonomous proliferation, but the safety of the approach remains to be established in patients.

LCK overexpressing CG1.CAR T cells showed superior IFN- γ release in response to low concentrations of the CG1 peptide, which is consistent with superior functional avidity. However, the superior antitumor effect of LCK overexpressing CG1.CAR T cells was more evident in in vivo experiments in which we also observed an enrichment of T stem memory–like T cells, lower frequency of T cells coexpressing TIM3 and PD1, and lower expression of TOX in T cells, which may suggest a less exhausted phenotype that requires additional proteomic and transcriptomic analysis. However, this observation suggests that LCK overexpression may promote a specific transcriptional program in CAR T cells. The transcriptome analysis revealed that LCK overexpression caused increased expression of genes encoding mitochondrial electron transport chain components after antigen stimulation. Functional investigation demonstrated a significant metabolic plasticity of LCK overexpressing CAR T cells and an increase in the mitochondrial biogenesis, which are all factors that have been correlated with T-cell memory⁵⁷ and superior fitness of CAR T cells and tumor-specific T cells in general.^{48,49,58–62} Additional investigation is needed to dissect the molecular link between LCK overexpression and mitochondrial biogenesis.

In summary, CAR T cells targeting an HLA-A2–restricted CG peptide demonstrate antitumor activity in AML without causing toxicity. LCK overexpression in CG1.CAR T cells not only promotes high functional avidity but promotes metabolic reprogramming of CAR T cells via improved mitochondrial mass.

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Authorship

Contribution: T.W. designed and performed most of the original experiments, analyzed data, and wrote the initial manuscript; F.S. designed and performed the experiments required for revision, analyzed the data, and generated the final figures; M.B. assisted with experiments, analyzed the data, and reviewed the manuscript; M.S. provided critical samples and supervised the experiments performed at Fondazione IRCCS San Gerardo dei Tintori; S.S., P.S., K.S., F.S., and R.T. assisted with the experiments; G.L. and L.S. performed the experiments in humanized mice; E.I.L. and M.W. performed the bioinformatics analysis; Z.Y., G.X., Z.C., P.W., and C.L. are employees of Eureka and performed the phage panning and screening; E.G.H., S.S., A.K., and J.E.T. assisted, performed, and analyzed the metabolic studies; L.F. prepared the libraries for RNA sequencing analysis; L.E.H., C.G.E., S.A.H., and P.M.A. performed the mass spectrometry studies; T.W., B.S., and P.M.A. conceived the study; G.D. and B.S. oversaw and directed the

research; G.D., F.S., and T.W. wrote the revised manuscript and the response to reviewers; and G.D. obtained research funding and wrote the manuscript.

Conflict-of-interest disclosure: G.D. serves in the scientific advisory board of NanoCells Therapeutics, Estella, Arovela, and Outpace Bio; and is cofounder of Persistence Bio. Z.Y., G.X., Z.C., P.W., and C.L. are employees of Eureka. The remaining authors declare no competing financial interests.

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Footnotes

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*F.S. and G.D. are joint last authors.

All RNA sequencing data have been uploaded to the Gene Expression Omnibus database (accession number GSE300381).

Raw data are available from authors Feifei Song (songff@med.unc.edu) and Gianpietro Dotti (gianpi@email.unc.edu) on request.

The online version of this article contains a data supplement.

There is a *Blood* Commentary on this article in this issue.

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